Michael Amir

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Professional Experience

Google DeepMind **London, UK**
Senior Research Scientist *2026–*
Working on provenance (SynthID).

Prorok Lab at University of Cambridge **Cambridge, UK**
Research Associate *2024–2026*
Developed methods for remotely verifying which policy a robot is running, and what it is trying to do, from observations of its motion. Also worked on hybrid methods where neural networks help improve, rather than replace, algorithms with formal guarantees.

Technion **Haifa, Israel**
Postdoctoral Fellow *2023*
Studied how agent failures and adversarial conditions affect the robustness of distributed robotic systems. Developed swarm algorithms that complete their mission while adapting to crashes and other unexpected conditions.

Education

Technion **Haifa, Israel**
PhD in Computer Science *2018–2023*
Thesis: "Multi-A(ge)nt Systems on Graphs"

Technion **Haifa, Israel**
MSc. in Computer Science *2015–2017*
Thesis: "Probabilistic Pursuits on Graphs"

The Open University **Israel**
B.A. in Mathematics and Philosophy *2009–2015*

Awards

Postdoctoral Affiliate of Trinity College, University of Cambridge · Runner-up for Rothschild Fellowship · National Excellence Award in Smart Transportation Research · Technion Research Excellence Award · Technion Best Research Poster Award · Technion Excellence Scholarship

Academic Mentorship

Samuel Ratnam · Hao Xiang Li · Tali Motzkin · Alon Shats · Mario Gomez Andreu · Lorenzo Magnino · Benjamin Chang · Sophie Chiang

Academic Activities

Invited Presentations (Selected)

March 2025: Alan Turing Institute: "Learning Dynamic Constraints for Multi-Agent Controllers"

September 2023: MIT Media Lab: "Natural Algorithms"

Peer Review / Program Committees

Nature Comms. Eng. · IEEE T-RO · ICML · ICLR · NeurIPS (Top Reviewer)
JAAMAS · JAIR · ECAI · CoRL · MRS · IROS · SAC · DARS · RSS

Publications

Journal Articles

Michael Amir and Alfred M. Bruckstein. "Time, Travel, and Energy in the Uniform Dispersion Problem". *IEEE Transactions on Robotics* (2025).

Dmitry Rabinovich*, **Michael Amir***, and Alfred M. Bruckstein. "Optimally Reordering Mobile Agents on Parallel Rows". *Theoretical Computer Science* 985 (2024).

Michael Amir, Noa Agmon, and Alfred M. Bruckstein. "A Locust-Inspired Model of Collective Marching on Rings". *Entropy* 24.7 (2022), 918.

Michael Amir and Alfred M. Bruckstein. "Probabilistic pursuits on graphs". *Theoretical Computer Science* 795 (2020), 459–477.

Conference Proceedings

Mario Gomez Andreu, **Michael Amir**, Antonio Marino, Tianxu An, Marco Hutter, and Amanda Prorok. "Leave It Better Than Found: Value-Guided Cooperation in Sequential Tasks". **Spotlight Paper** at CoRL 2026.

Mae Frost, **Michael Amir**, and Shaunak D. Bopardikar. "Games Over Observation Spaces in Multi-Agent Capture the Flag". *Accepted to GameSec 2026*. arXiv: 2609.06178 [cs.GT].

Rishabh Jain, Keisuke Okumura, **Michael Amir**, and Amanda Prorok. "Graph Attention-Guided Search for Dense Multi-Agent Pathfinding". **Oral Paper** at AAAI 2026. arXiv: 2510.17382 [cs.AI].

Michael Amir*, Matteo Bettini*, and Amanda Prorok. "When Is Diversity Rewarded in Cooperative Multi-Agent Learning?". *Accepted to ICLR 2026*. arXiv: 2506.09434 [cs.MA].

Michael Amir*, Manon Flageat*, and Amanda Prorok. "Remotely Detectable Robot Policy Watermarking". *Accepted to ICLR 2026*. arXiv: 2512.15379 [cs.RO].

Rishabh Jain, Keisuke Okumura, **Michael Amir**, Pietro Lio, and Amanda Prorok. "Pairwise is Not Enough: Hypergraph Neural Networks for Multi-Agent Pathfinding". *Accepted to ICLR 2026.*

Hao Xiang Li, **Michael Amir**, and Amanda Prorok. "Scaling Multi-Agent Environment Co-Design with Diffusion Models". *Accepted to ICML 2026*. arXiv: 2511.03100 [cs.LG].

Michael Amir, Guang Yang, Zhan Gao, Keisuke Okumura, Heedo Woo, and Amanda Prorok. "ReCoDe: Reinforcement Learning-based Dynamic Constraint Design for Multi-Agent Coordination". Proceedings of 2025 Conference on Robot Learning (CoRL2025). 2025. arXiv: 2507.19151 [cs.RO].

Michael Amir*, Dmitry Rabinovich*, and Alfred M. Bruckstein. "Patrolling Grids with a Bit of Memory". *Proc. of the Intl. Workshop on the Algorithmic Foundations of Robotics (WAFR)*. Chicago, USA, 2024.

Michael Amir, Yigal Koifman, Yakov Bloch, Ariel Barel, and Alfred M. Bruckstein. "Multi-Agent Distributed and Decentralized Geometric Task Allocation". *Proc. of the IEEE Conference on Decision and Control (CDC)*. 2023.

Ori Rappel*, **Michael Amir***, and Alfred M. Bruckstein. "Stigmergy-based, Dual-Layer Coverage of Unknown Indoor Regions". *Proc. of the Intl. Conf. on Autonomous Agents and MultiAgent Systems (AAMAS)*. London, United Kingdom, 2023.

Alon Shats, **Michael Amir**, and Noa Agmon. "Competitive Ant Coverage: The Value of Pursuit". *Proc. of the IEEE/RSJ Intl. Conf. on Intelligent Robots and Systems (IROS)*. Detroit, United States, 2023.

Michael Amir, Noa Agmon, and Alfred M. Bruckstein. "A Discrete Model of Collective Marching on Rings". *Proc. of the Intl. Symposium on Autonomous Robotic Systems (DARS)*. 2021.

Michael Amir and Alfred M. Bruckstein. "Fast Uniform Dispersion of a Crash-Prone Swarm". *Robotics: Science and Systems (RSS)*. 2020.

Michael Amir and Alfred M. Bruckstein. "Minimizing Travel in the Uniform Dispersal Problem for Robotic Sensors". *Proc. of the Intl. Conf. on Distributed Autonomous Agents and MultiAgent Systems (AAMAS)*. Montreal, Canada, 2019.

Peer-reviewed Workshops

Mateusz Odrowaz-Sypniewski, **Michael Amir**, Jasmine Bayrooti, Ajay Shankar, and Amanda Prorok. "Learning Robot Behavior Priors from Non-Robotic Synthetic Data". *Accepted to the 8th Robot Learning Workshop: Is Physical AI Going Zero-Shot? (NeurIPS 2026)*.

Rishabh Jain, Keisuke Okumura, **Michael Amir**, Pietro Lio, and Amanda Prorok. "Pairwise is Not Enough: Hypergraph Neural Networks for Multi-Agent Pathfinding". **Oral Paper** at the AAAI 2026 Workshop on Multi-Agent Path Finding (WoMAPF).

Preprints

Benjamin Chang, **Michael Amir**, Manon Flageat, and Amanda Prorok. *Remotely Detectable Keyed Communication through Motion*. 2026. arXiv: 2609.08920 [cs.RO].

Eleanor Clifford*, **Michael Amir***, Arduin Findeis, Aaron Zhao, and Robert Mullins. *Open Problems in Constitutional Preference Reconstruction*. 2026. arXiv: 2606.30116 [cs.AI].

Rishabh Jain, Akmaral Moldagalieva, Lorenzo Magnino, **Michael Amir**, Keisuke Okumura, Ajay Shankar, Wolfgang Hönig, and Amanda Prorok. *GAMBIT: Learning to Plan Continuous Multi-Robot Trajectories*. 2026. arXiv: 2610.06290 [cs.RO].